

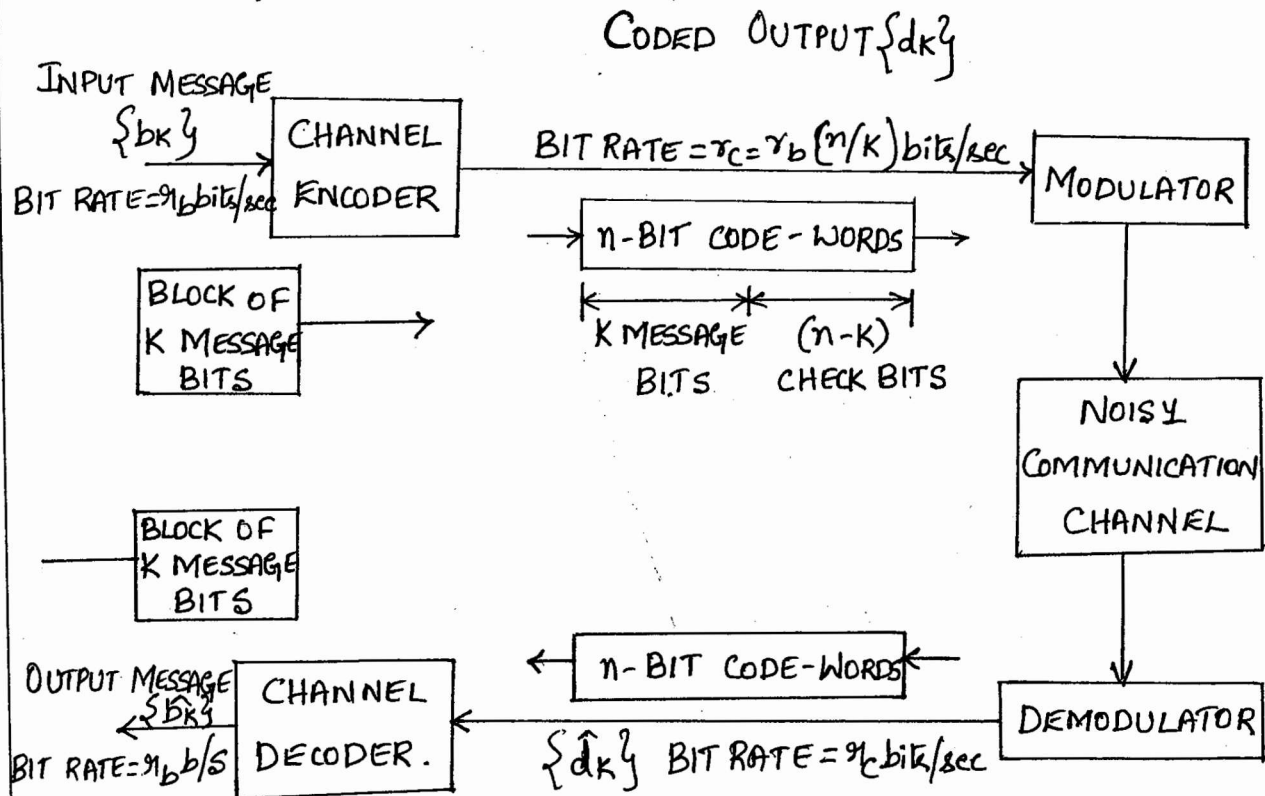
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ERROR CONTROL CODING.

(14)

Block diagram of Communication System Employing error control Coding.



The above block diagram gives us a digital communication system, employing error control coding. The main functional blocks are the channel encoder, the channel decoder, modulator and demodulator & the noisy communication channel with a Capacity C bits/sec.

The source generates a message block $\{b_k\}$ at a rate of γ_b bits/sec and feeds it to the channel encoder.

The channel encoder, then adds $(n-k)$ number of redundant bits to these k -bit messages to form n -bit code words. These $(n-k)$ number of additional bits also called "check bits" do not carry any information but helps channel decoder to detect & correct errors.

The bit rate of the coded output block $\{d_k\}$ will be $r_c = r_b(n/k)$ bits/sec. This is the rate at which the modem operates, to produce a message block $\{\hat{d}_k\}$ at the receiver. The channel decoder then decodes this message to get back the information block $\{\hat{b}_k\}$ at the receiver.

The information block $\{\hat{b}_k\}$ usually differs from the transmitted block $\{b_k\}$. Thus probability of error.

$$P_e = P\{\hat{b}_k \neq b_k\}$$

then the probability of error which depends on bit rate r_c is defined as

$$r_c = P\{\hat{d}_k \neq d_k\}$$

$\therefore$ It is required to design an error control coding scheme such that the overall probability of error is less than the desired value.

Methods of Controlling Errors.

1. Forward acting error correction : In this method the errors are detected and corrected by proper coding techniques. The check bits/redundant bits are used to detect & correct errors. The error detection & correction capability depends on the number of redundant bits in the transmitted message. The FAEC is faster but the probability of error is comparatively high.
2. Error Detection : In this method, the decoder checks the i/p sequence, when it detects any error it rejects that part of the sequence & request the Tx^r for retransmission. hence the decoder doesnot correct the errors. It just detects the errors & sends request to the Tx^r through a reverse channel for retransmission. this method has low probability of error but high

Types of Errors :

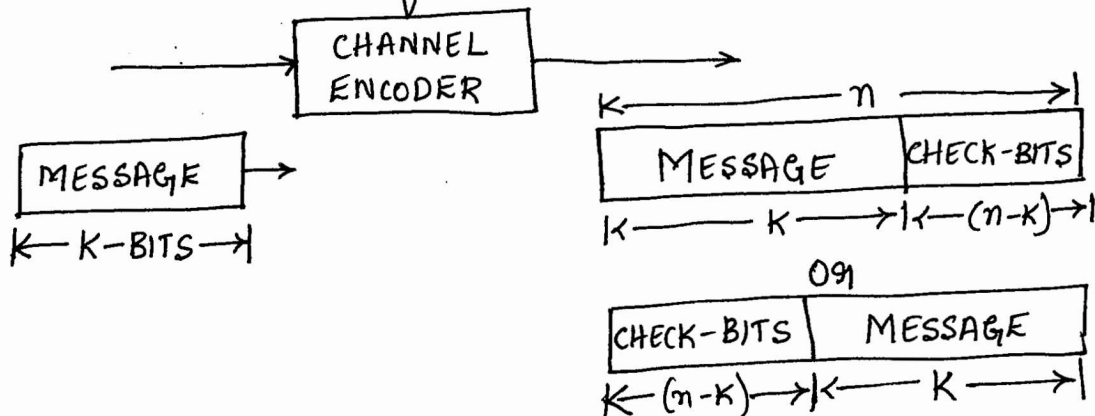
- (i) Random Error : The transmission errors that occur due to the presence of white gaussian noise are referred to as "random errors". Sources of Gaussian noise include thermal & shot noise in the transmitting and receiving equipment.
- (ii) Burst error : These errors are generated by Impulsive noise in the channel, this noise is due to lightning & switching transients & man-made noise etc. This noise affect more than one symbol and the error caused is called "Burst Error".

Types of Codes :

- (i) Block Codes : These codes consists of 'n' number of bits in one block or codeword. This codeword consists of 'K' message bits and (n-K) redundant bits such block codes are called (n-K) block codes.
- (ii) Convolution Codes : In this code, the check bits are continuously interleaved with information bits. The check bits verify the information bits not only in the block immediately preceding them, but in other blocks as well.

LINEAR BLOCK CODES.

- Formation of Linear Block Codes.



In channel encoder, a block of 'k' message bits is encoded into a block of 'n' bits by adding $(n-k)$ number of check bits as shown in figure.

where $n > k$, such a code formed is called (n, k) block code. A (n, k) block code is said to be a " (n, k) linear block code" if it satisfies the condition:

Let C_1 & C_2 be any 2 code-words belonging to a set of (n, k) block code. if $C_1 \oplus C_2$ is also a n -bit code-word belonging to the same set of (n, k) block code, then such a block code is called (n, k) linear block code.

A (n, k) linear block code is said to be "systematic" if the k -message bit appear either at the "beginning" of the code-word or at the end of the code-word as shown in figure.

Matrix Description of linear Block codes.

The encoding operation in a linear block encoding scheme consists of 2 basic steps.

- ① The information sequence is segmented into message blocks each block consisting of 'k' successive information bits.
- ② The encoder transforms each message block into a larger block of n bits according to some predetermined set of rules.

These $n-k$ additional bits are generated from linear combinations of the message bits, & we can describe the encoding operations using matrices.

- Let the message block of k -bits be represented as a "row-vector" or " k -tuple" called "message-vector" given by

$$[D] = \{d_1, d_2, d_3, \dots, d_k\} \longrightarrow \text{①}$$

where $d_1, d_2, \dots, d_k$ are either '0's or '1's.

$\therefore$ There are 2^k distinct message vectors.

The channel encoder systematically adds $(n-k)$ number of check bits to form a (n, k) linear block code. Then the 2^k code vectors can be represented by

$$C = \{c_1, c_2, \dots, c_n\} \rightarrow (2)$$

The ratio (k/n) is defined as the "rate efficiency" of the (n, k) linear block code.

In a systematic linear block code, the message bits appear at the beginning of the code-vector or at the end of the code vector

$$\therefore c_i = d_i \text{ for all } i=1, 2, \dots, k \rightarrow (3)$$

The remaining $(n-k)$ bits are check bits, hence eqnⁿ (2) & (3) can be combined as

$$[C] = \underbrace{\{c_1, c_2, \dots, c_k\}}_{k\text{-message bits}}, \underbrace{\{c_{k+1}, c_{k+2}, \dots, c_n\}}_{(n-k)\text{ check bits}} \rightarrow (4)$$

These $(n-k)$ check bits $c_{k+1}, c_{k+2}, \dots, c_n$ are derived from k -message bits using a predetermined rule

$$\left. \begin{aligned} \text{i.e. } c_{k+1} &= p_{11}d_1 + p_{21}d_2 + \dots + p_{k1}d_k \\ c_{k+2} &= p_{12}d_1 + p_{22}d_2 + \dots + p_{k2}d_k \\ &\vdots \\ c_n &= p_{1, n-k}d_1 + p_{2, n-k}d_2 + \dots + p_{k, n-k}d_k \end{aligned} \right\} \rightarrow (5)$$

where $p_{11}, p_{21}, p_{12}, p_{22}, \dots$ are either '0's & '1's and the addition above is performed using modulo-2 arithmetic.

$\therefore$ combining (3), (4) & (5) we can express the result in matrix form as

$$[c_1 \ c_2 \ \dots \ c_k \ c_{k+1} \ c_{k+2} \ \dots \ c_n] = [d_1 \ d_2 \ \dots \ d_k] \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & p_{11} & p_{12} & \dots & p_{1n-k} \\ 0 & 1 & \dots & 0 & p_{21} & p_{22} & \dots & p_{2n-k} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & p_{k1} & p_{k2} & \dots & p_{kn-k} \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{k\text{-terms}} \quad \underbrace{\hspace{10em}}_{(n-k)\text{terms}}$

$$[c] = [p][G]$$

where $[G]$ is called as "Generator Matrix" of order $(k \times n)$ given by

$$[G] = [I_k \mid P]_{k \times n}$$

where I_k = unit matrix of order 'k'

$[P]$ = arbitrary matrix called "PARITY MATRIX" of order $k \times (n-k)$

& $\mid$ = denotes the demarcation b/w the unit matrix I_k & parity matrix P .

The parity 'P' is selected to correct random & burst errors.

Modulo-2 arithmetic

Modulo-2 addition	Modulo-2 Multiplication	Modulo-2 Subtraction
$0 + 0 = 0$	$0 * 0 = 0$	$0 - 0 = 0$
$0 + 1 = 1$	$0 \cdot 1 = 0$	$0 - 1 = 1$
$1 + 0 = 1$	$1 \cdot 0 = 0$	$1 - 0 = 1$
$1 + 1 = 0$	$1 \cdot 1 = 1$	$1 - 1 = 0$

PROBLEMS.

- (1) For a systematic $(6,3)$ linear Block code, the parity matrix P is given by $[P] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ find all possible code-vectors.

→ Solution.

We know (n, K) linear block codes

Given: $(6,3)$ — " —

$$\therefore n=6, K=3$$

There are 2^K distinct message vectors $\Rightarrow 2^3 = 8$
 $(000), (001), (010), (011), (100), (101), (110), (111)$

The code vectors are then found by using eqn

$$[C] = [D] [G]$$

$$\text{where } [G] = [I_K | P]$$

$$= [I_3 | P]$$

$$\therefore [G] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\therefore [C] = [D] [G] = [d_1 \ d_2 \ d_3] \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$= [d_1, d_2, d_3, (d_1+d_3), (d_2+d_3), (d_1+d_2)]$$

∴ The code vectors can be found which are listed

Code name	Message-vector	Code-vector for (6,3) linear block code
C_1	000	000000
C_2	001	001110
C_3	010	010011
C_4	011	011101
C_5	100	100101
C_6	101	101011
C_7	110	110110
C_8	111	111000

To verify it is linear block code

$$C_1 \oplus C_7 = 000000$$

$$\begin{array}{r} 110110 \\ \hline 110110 \end{array}$$

$\Rightarrow C_7$ which is present in the block code

hence (6,3) is a linear block code.

2. For a systematic (7,4) linear Block code whose Generator Matrix is given (i) find all the Code words.

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

→ Given $(n, k) = (7, 4)$ linear Block codes

∴ $2^k = 2^4$ message-vectors . i.e 16 msg vectors

code vectors C_1 to C_{16}

message vectors 1 to 16.

Code vectors are found by using $[c] = [D] [G]$

$$[C] = [d_1 \ d_2 \ d_3 \ d_4] \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$[c] = [d_1, d_2, d_3, d_4, (d_1 + d_2 + d_3), (d_2 + d_3 + d_4), (d_1 + d_2 + d_4)]$$

The code-vectors are listed below

Code name	Message-vector	Code-vector
C_1	0000	00000000
C_2	0001	00010111
C_3	0010	00101110
C_4	0011	00111011
C_5	0100	01001111
C_6	0101	01011000
C_7	0110	01100101
C_8	0111	01110010
C_9	1000	10001011
C_{10}	1001	10010111
C_{11}	1010	10101110
C_{12}	1011	10111000
C_{13}	1100	11000110
C_{14}	1101	11010011
C_{15}	1110	11101000
C_{16}	1111	11111111

Parity check matrix

we have Generator matrix given by

$$[G] = [I_k | P]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \dots & 0 & P_{11} & P_{12} & \dots & P_{1,n-k} \\ 0 & 1 & 0 & \dots & 0 & P_{21} & P_{22} & \dots & P_{2,n-k} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 1 & P_{k1} & P_{k2} & \dots & P_{k,n-k} \end{array} \right] \rightarrow (1)$$

$\underbrace{\hspace{10em}}_{k \text{ columns}} \quad \underbrace{\hspace{10em}}_{(n-k) \text{ columns}}$

The generator matrix $[G]$ having another matrix called "Parity check matrix - H" given by

$$[H] = [P^T | I_{n-k}]$$

$$[H] = \left[\begin{array}{ccc|ccc} P_{11} & P_{12} & \dots & P_{1k} & 1 & 0 & 0 & \dots & 0 \\ P_{21} & P_{22} & \dots & P_{2k} & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{1(n-k)} & P_{2(n-k)} & \dots & P_{k(n-k)} & 0 & \dots & 0 & \dots & 1 \end{array} \right]$$

$\underbrace{\hspace{10em}}_{k \text{ columns}} \quad \underbrace{\hspace{10em}}_{(n-k) \text{ columns}}$

$\therefore [H]$ matrix is a $(n-k) \times (n)$ matrix.

Encoding circuit for (n,k) linear block codes:

we know/have the matrix form as

$$[C_1, C_2, C_3, \dots, C_k, C_{k+1}, C_{k+2}, \dots, C_n] = [d_1, d_2, \dots, d_k] \left[\begin{array}{ccc|ccc} \underbrace{1 \ 0 \ 0 \ 0 \dots 0}_{k \text{ terms}} & \underbrace{P_{11} \ P_{12} \dots P_{1,n-k}}_{(n-k) \text{ terms}} \\ 0 \ 1 \ 0 \ 0 \dots 0 & P_{21} \ P_{22} \dots P_{2,n-k} \\ \vdots & \vdots \\ 0 \ 0 \ 0 \ 0 \dots 1 & P_{k1} \ P_{k2} \dots P_{k,n-k} \end{array} \right]$$

$\underbrace{\hspace{10em}}_{k \times n} \quad (1)$

Then

$$C_1 = d_1$$

$$C_2 = d_2$$

$\vdots$

$$C_K = d_K$$

$$C_{K+1} = P_{11}d_1 + P_{21}d_2 + \dots + P_{K1}d_K$$

$$C_{K+2} = P_{12}d_1 + P_{22}d_2 + \dots + P_{K2}d_K$$

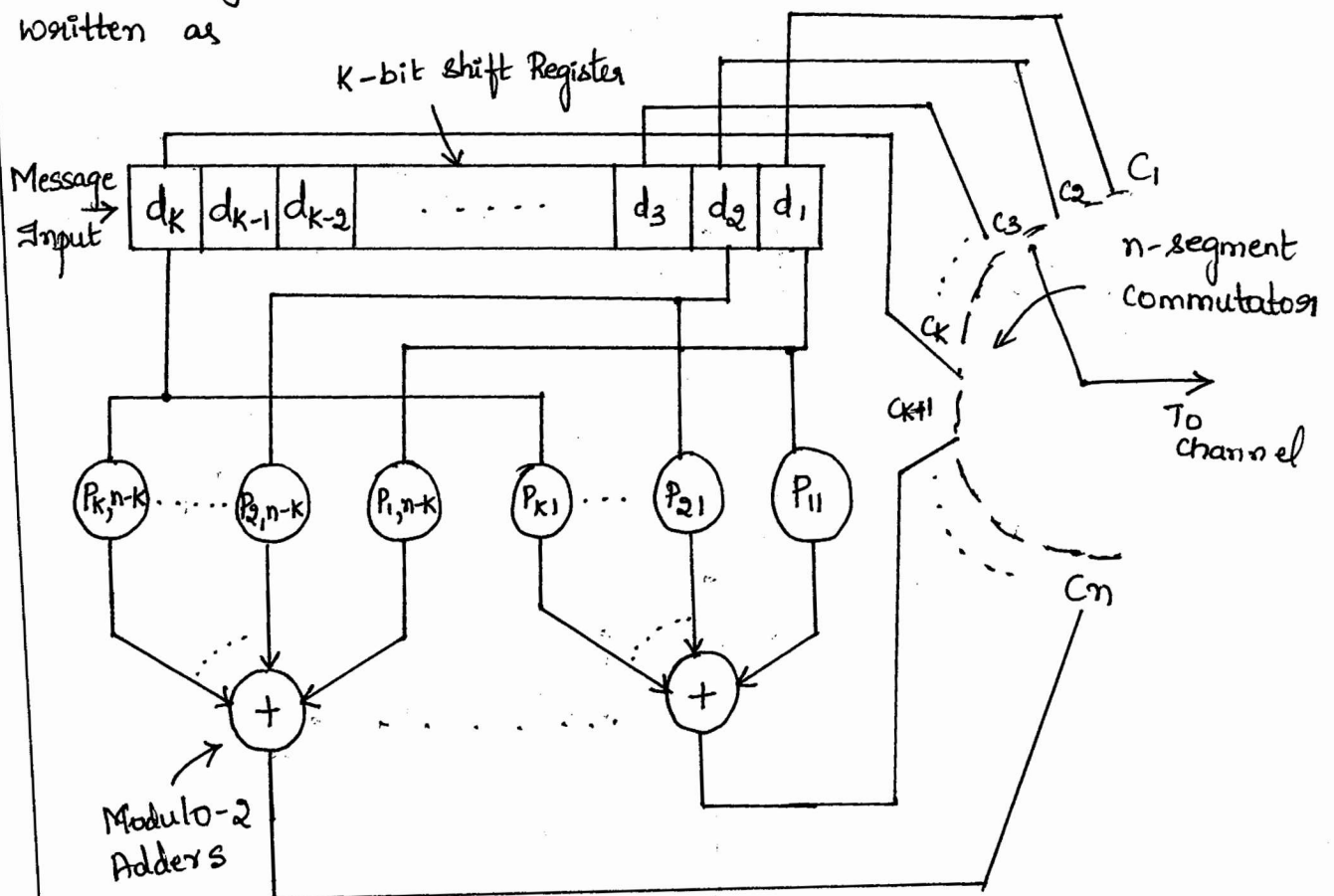
$\vdots$

$$C_n = P_{1,n-K}d_1 + P_{2,n-K}d_2 + \dots + P_{K,n-K}d_K$$

→ (2)

The design of encoder circuit consists of a K-bit shift register, a n-segment commutator & (n-K) number of modulo-2 adders.

∴ Encoding circuit for (n,k) linear block codes can be written as



The entire data $d_k, d_{k-1}, d_{k-2}, \dots, d_2, d_1$ is shifted into the k -bit shift register. The small circles $P_{11}, P_{21}, \dots, P_{k1}$ are either "open-circuit" or "short-circuit" depending on either '0' or '1' ex:- if P_{11} is 0 then there is no connection from d_1 to the modulo-2 adder and if $P_{11}=1$, then there is connection.

When the message is shifted into the shift register, the modulo-2 adders generate the 'check bits' which are fed into the commutator segments along with message bit as shown in figure.

problem

1) Construct the encoding circuit for problem (1) in pg (4)

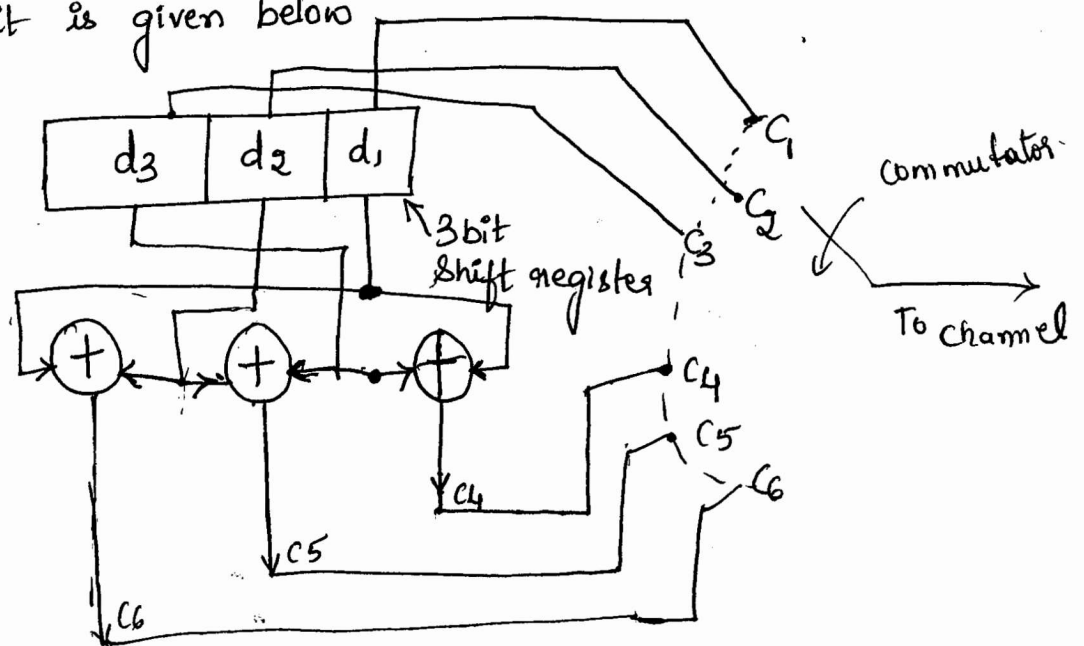
→ The code-vectors are given by

$C_1 = d_1, C_2 = d_2, C_3 = d_3, C_4 = d_1 + d_3, C_5 = d_2 + d_3, C_6 = d_1 + d_2$
we require 3 shift registers.

where we have (n, k) linear codes
 $(6, 3)$ — " —

$n - k = 6 - 3 = 3$, we require 3 modulo adders & 6 segment commutator.

the circuit is given below



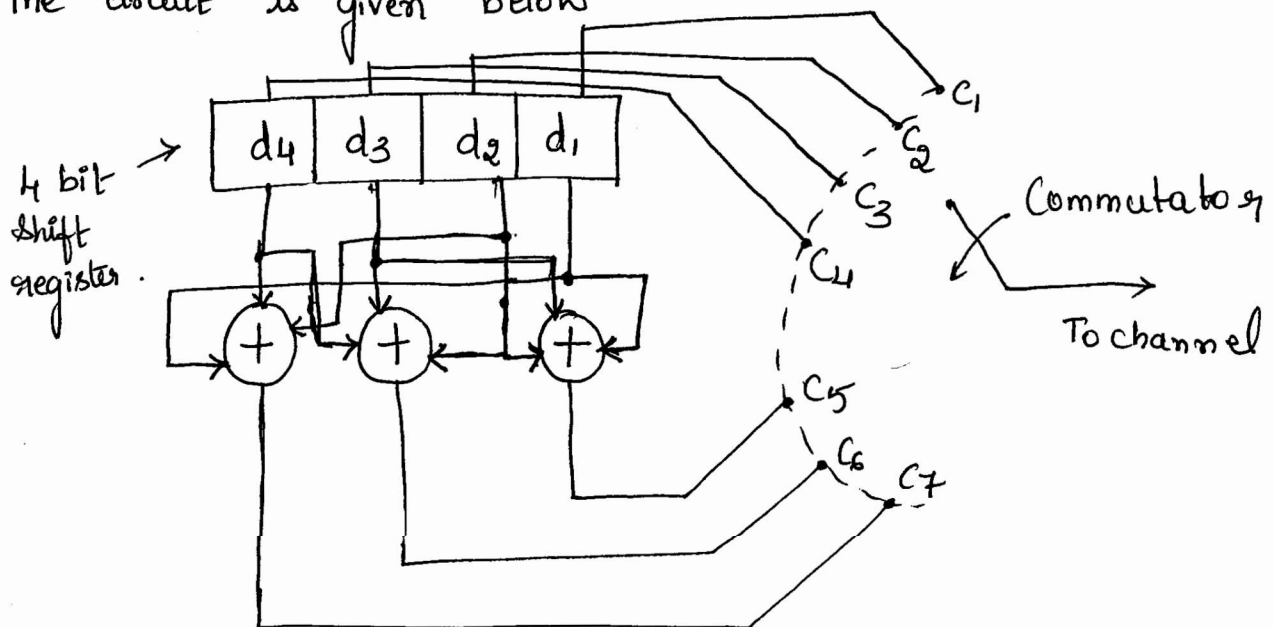
2) Construct the encoding circuit for problem (2) in pg (4)
 → the code vectors are given by

$$C_1 = d_1, C_2 = d_2, C_3 = d_3, C_4 = d_4, C_5 = (d_1 + d_2 + d_3), \\ C_6 = (d_2 + d_3 + d_4), C_7 = (d_1 + d_2 + d_4)$$

we have (7,4) linear block code

we require 4 shift registers, 7 segment Commutator,
 $n-k = 6-3 = 3$, we require 3 modulo adders

The circuit is given below



(3) If C is a Valid Code Vector, then prove that $CH^T = 0$
 where H^T is the transpose of the parity check matrix H .

→ we have $[G] = [I_k | P] = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & P_{11} & P_{12} & \dots & P_{1,n-k} \\ 0 & 1 & 0 & \dots & 0 & P_{21} & P_{22} & \dots & P_{2,n-k} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots & 0 & P_{k1} & P_{k2} & \dots & P_{k,n-k} \end{bmatrix}$

∴ The i th row of the $[G]$ matrix is given by

$$g_i = \left[0 \ 0 \ \dots \ 1 \ \dots \ 0 \ P_{i1} \ P_{i2} \ \dots \ P_{ij} \ \dots \ P_{i,(n-k)} \right]$$

$\uparrow$
 i th element

$\uparrow$
 $(K+j)$ th elements (1)

we have $[H] = [P^T | I_{n-k}]$

$$\therefore [H] = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1k} & 1 & 0 & 0 & \dots & 0 \\ P_{21} & P_{22} & \dots & P_{2k} & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{ij} & P_{2j} & \dots & P_{kj} & 0 & 0 & 0 & \dots & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{1, n-k} & P_{2, n-k} & \dots & P_{k, n-k} & 0 & 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$h_j = [P_{1j} \ P_{2j} \ \dots \ P_{ij} \ \dots \ P_{kj} \ 0 \ 0 \ \dots \ 1 \ \dots \ 0] \rightarrow (2)$$

$\uparrow$ i^{th} element $\uparrow$ $(K+j)^{\text{th}}$ element

Let us consider $g_i h_j^T = [0 \ 0 \ \dots \ 1 \ \dots \ 0 \ P_{1i} \ P_{2i} \ \dots \ P_{ij} \ \dots \ P_{i, n-k}]$

$\uparrow$ i^{th} element

$\uparrow$ $(K+j)^{\text{th}}$ element

P_{1j}
 P_{2j}
 $\vdots$
 P_{ij}
 $\vdots$
 P_{kj}
 $\vdots$
 0
 $\vdots$
 1
 $\vdots$
 0

$\rightarrow (3)$

Modulo-2 multiplication of the row and column yields

$$g_i h_j^T = (0)(P_{1j}) + (0)(P_{2j}) + \dots + (1)(P_{ij}) + \dots + 0(P_{kj}) + \dots + P_{1i}(0) + P_{2i}(0) + \dots + P_{ij}(1) + \dots + P_{i, n-k}(0)$$

$$g_i h_j^T = P_{ij} + P_{ij} = P_{ij}(1+1) = 0 \rightarrow (4) \text{ this equn (4) is true for every values of } i \text{ \& } j$$

$$\therefore \text{we have } [G][H]^T = 0 \rightarrow (5)$$

pre-multiplying $[D]$ in both sides of equation (5), we get

$$[D][G][H]^T = [D][0]$$

But we have $[C] = [D][G]$

$$\therefore [C][H]^T = 0$$

$$[CH^T = 0]$$

Syndrome and Error Correction.

Consider the transmitted code-vector $[C] = [c_1, c_2, c_3 \dots c_n]$ and received code vector $R = [r_1, r_2, \dots r_n]$

If $R = C$, then there is no error

If $R \neq C$, then there is an error in received vector.

$\therefore$ The error vector 'e' can be represented as a vector by $E = (e_1, e_2, e_3 \dots e_n)$ & it is defined as the difference b/w 'R' and 'C'

$$\therefore E = R - C = R + C \rightarrow \textcircled{1} \quad (\because \text{add \& sub is same in modulo-2 arithmetic})$$

If $r_i = c_i$, $e_i = 0$, if $r_i \neq c_i$, $e_i = 1$

The Receiver Knows only 'R' & does not know 'C' and 'E'.

In order to detect the errors in the received vectors we define an $(n-k)$ bit vector known as the error syndrome of the circuit. It is defined as

$$[S] = [R][H]^T \rightarrow \textcircled{2}$$

But we know $R = C + E$ from $\textcircled{1} \rightarrow \textcircled{A}$ for $S = (s_1, s_2 \dots s_{n-k})$

$$\therefore S = [C + E][H]^T$$

$$= CH^T + EH^T$$

$$S = EH^T \rightarrow \textcircled{3}$$

So from equⁿ $\textcircled{3}$, the receiver finds 'E' where S & H^T are known.

—|| —|| $\textcircled{A}$, the code-vector 'C' can be found out

If $S = 0$, then the received vector is same as code vector
i.e there is no error

$S \neq 0$, then an error has occurred in R.

Problems.

1. Consider $(7,4)$ LBC with the parity matrix as $P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(i) find $[G], [H]$

(ii) If there are 3 received vectors as

$$R_1 = 0100000, R_2 = 1110111, R_3 = 1100100$$

determine if there was an error during transmission, also detect & correct the errors.

Solⁿ

(i) $[G] = [I_k \mid P]_{k \times n}$

we have $(n, k) = (7, 4)$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}_{4 \times 7}$$

$$[H] = [P^T \mid I_{n-k}]$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 7}$$

(i) $[S] = RH^T$

the given first $R_1 = 0100000 = [0100000] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{7 \times 3}$

$$S_1 = [1 \ 0 \ 1]$$

$\therefore S_1 \neq 0$ There is an error

Match S_1 with rows of H^T , In H^T , 2 row matches with:
 $\therefore$ The 2nd bit of R_1 is in error and corrected code vector is $\hat{C}_1 = 0000000$

(ii) given $S_2 = [110111]$ $R_2 = [111011]$

$$S_2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S_2 = [1 \ 1 \ 1]$$

$\therefore S_2 \neq 0 \Rightarrow$ There is an error
 Match S_2 with rows of H^T , In H^T 4th row matches with S_2
 $\therefore$ the 4th bit of R_2 is in error & corrected code vector is
 $\hat{C}_2 = [111111]$

(iii) given $S_3 = [1100100]$ $R_3 = [1100100]$

$$S_3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$S_3 = [0 \ 1 \ 0]$
 $\therefore S_3 \neq 0$ there is an error $\Rightarrow$ 11th as above $\hat{C}_3 = [1100100]$
 corrected vector

(2) For (6,3) linear block code the 'G' is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

find (i) H

(ii) If $C = 111000$ find 'syndrome' if $R = 111001$ & determine the transmitted code word.

$$\rightarrow (i) H = [P^T \mid I_{n-k}]$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore H^T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(ii) S = CH^T$$

$$111000 \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since $S=0$, the received bit is same as transmitted bit

Given $R = 111001$

$$S = RH^T \Rightarrow [111001] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = [0 \ 0 \ 1]$$

The error occurs in the 6th row, this means that there is an error in 6th bit of message transmitted

$\therefore$ Transmitted code word is $[111000]$

Syndrome Calculation Circuit.

We know that $S = RH^T \rightarrow (1)$

Expanding (1) in general format

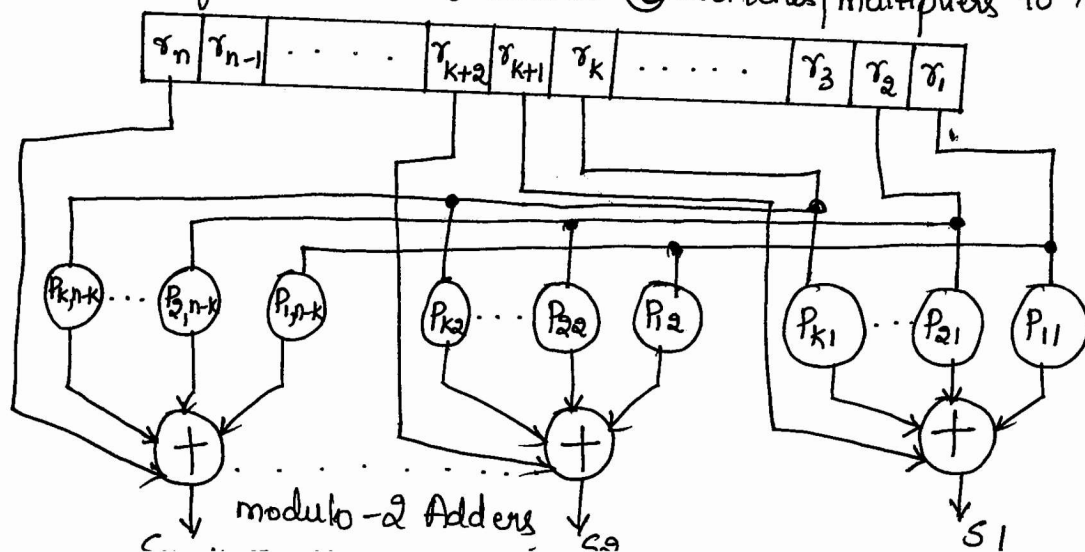
$$\therefore [S_1, S_2, \dots, S_{n-k}] = [r_1, r_2, \dots, r_n] \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1,n-k} \\ p_{21} & p_{22} & \dots & p_{2,n-k} \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1} & p_{k2} & \dots & p_{k,n-k} \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} \rightarrow (2)$$

Multiplying by using modulo-2 arithmetic, we get the syndrome bits as

$$\left. \begin{aligned} S_1 &= r_1 p_{11} + r_2 p_{21} + \dots + r_k p_{k1} + r_{k+1} \\ S_2 &= r_1 p_{12} + r_2 p_{22} + \dots + r_k p_{k2} + r_{k+2} \\ &\vdots \\ S_{n-k} &= r_1 p_{1,n-k} + r_2 p_{2,n-k} + \dots + r_k p_{k,n-k} + r_n \end{aligned} \right\} \rightarrow (3)$$

So from (3), the hardware requirement for the syndrome calculation circuit for (n,k) linear block code can be summarised as follows:

- A n -bit shift register to hold the received vector r
- Max of $(n-k) \bmod 2$ address (2) switches/multipliers to realize p_{ij} .



Problems -

- ① For the given parity matrix, construct the syndrome circuit.

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\rightarrow H = [P^T \mid I_{n-k}]$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}_{3 \times 6}$$

$$S = R H^T$$

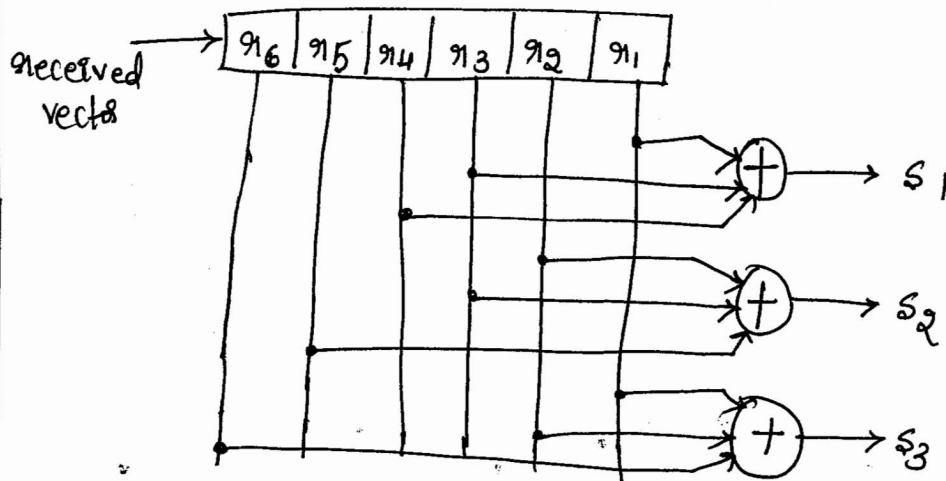
$$\therefore S \Rightarrow [R_1 R_2 R_3 R_4 R_5 R_6] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

let $R = R_1, R_2, R_3, R_4, R_5, R_6$

$$S_1 = r_1 + r_3 + r_4, \quad S_2 = r_2 + r_3 + r_5, \quad S_3 = r_1 + r_2 + r_6$$

$$\therefore n=6$$

$$n-k = \text{modulo adder} = 6-3 = 3 \text{ syndrome.}$$



- (2) For a given (7,4) linear block code, the parity matrix P is given by
- $$[P] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$
- ① find all possible valid code-vectors
 ② Draw encoding circuit
 ③ A single error has occurred in each of these received vectors. Detect and correct those errors
 (a) $R_A = [0111110]$ (b) $R_B = [1011100]$ (c) $R_C = [1010000]$

- (4) Draw the syndrome calculation circuit.

→ To find code-vectors, we have $[c] = [D][E]$

$$2^K = 2^4 = 16$$

Where $[E] = [I_K | P]$

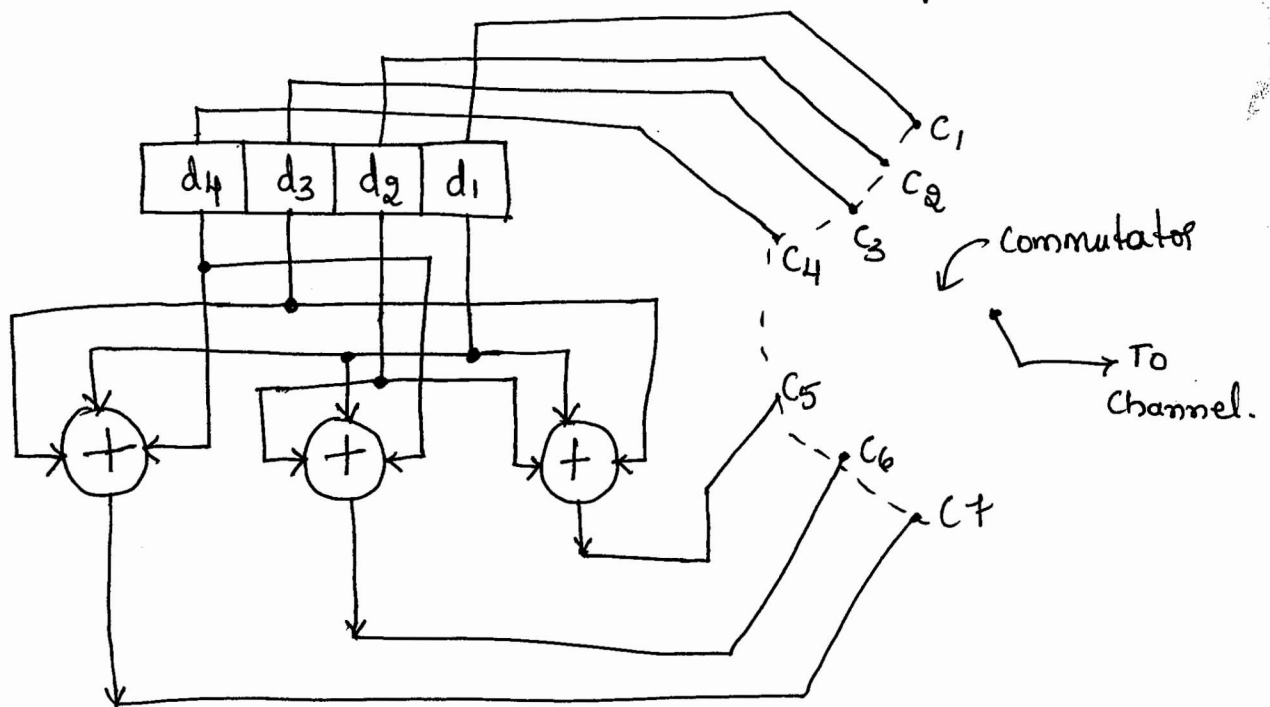
$$= \begin{bmatrix} 1 & 0 & 0 & 0 & | & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & | & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & | & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & | & 0 & 1 & 1 \end{bmatrix}$$

$$[c] = [D][E] \\ = [d_1 \ d_2 \ d_3 \ d_4] \begin{bmatrix} 1 & 0 & 0 & 0 & | & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & | & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & | & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & | & 0 & 1 & 1 \end{bmatrix}$$

$$[c] = [d_1, d_2, d_3, d_4, d_1+d_2+d_3, d_1+d_2+d_4, d_1+d_3+d_4]$$

Message Vector (D)	Code-Vectors (C)	Message Vector (D)	Code-Vectors (C)
0000	0000000	1000	1000111
0001	0001011	1001	1001100
0010	0010101	1010	1010010
0011	0011110	1011	1011001
0100	0100110	1100	1100001
0101	0101101	1101	1101010
0110	0110011	1110	1110100
0111	0111000	1111	1111011

- (ii) The encoding circuit which requires a 4-bit shift register, $(n-k) = (7-4) \Rightarrow 3 \text{ modulo-2 adders}$ & 7 segment Commutator



- (iii) To detect and correct errors

$$[S] = R H^T$$

$$[H] = [P^T \mid I_{n-k}] = [P^T \mid I_3] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Given $R_A = [0111110] \Rightarrow [S_1] = R_A H^T$

$$= [0111110] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore [S] = [1 \ 1 \ 0]$$

$\therefore [S_1] \neq 0$, In H^T the 2nd row matches with S_1 , $\therefore$ the 2nd bit of R_A is in error and corrected code vector is $C_A = 0011110$ which is valid code vector.

Given $R_B = [1011100]$

So $[S_2] = R_B H^T$

$$= [1011100] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[S_2] = [101]$$

$\therefore [S_2] \neq 0$, 3rd row matches with S_2 , $\therefore$ The corrected code vector $C_B = 1001100$

Given $R_C = [1010000]$

So $[S_3] = R_C H^T$

$$= [1010000] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[S_3] = [010]$$

$\therefore [S_3] \neq 0$, 6th row matches with S_3 , $\therefore$ The corrected code vector $C_C = 1010010$.

(iv) Syndrome Calculation Circuit.

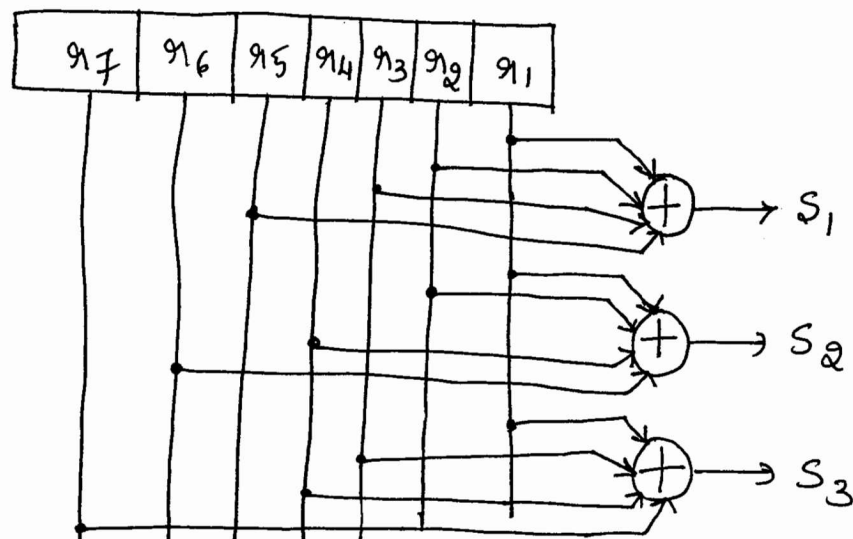
Let the received vector be represented in general form

$$R = [r_1, r_2, r_3, r_4, r_5, r_6, r_7]$$

The received vector R is

$$S = R H^T = [r_1, r_2, r_3, r_4, r_5, r_6, r_7] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore S = [S_1, S_2, S_3] = [(r_1 + r_2 + r_3 + r_5), (r_1 + r_2 + r_4 + r_6), (r_1 + r_3 + r_4 + r_7)]$$



Hamming Weight, Hamming distance & minimum distance of Linear block codes.

Hamming Weight: It is a measure of the total number of non-zero elements (1's) in a given code vector.

eg:- If $C_2 = 0110110 \rightarrow$ then $H_w = 4$. (i.e. no of 1's is 4)

Hamming Distance: Hamming distance between any two code-vectors C_1 and C_2 is defined as the number of components in which they differ.

eg:-

	1 st	2 nd	3 rd	4 th	5 th	6 th
$C_1 =$	0	0	1	0	1	1
$C_2 =$	1	0	0	1	1	0
	↑	↑	↑	↑		
	1 st	2 nd	3 rd	4 th		

The hamming distance between C_1 and C_2 is said to be 4.

Minimum Distance [d_{min}]: Minimum distance is defined as the Smallest hamming distance between any two code-vectors in a code.

d_{min} can also be measured by knowing all the hamming weights of a given code and $d_{min} = H_{w \min}$.

eg:- If $C_3 = 0110100 \rightarrow h_w = 3$

$C_4 = 1100110 \rightarrow h_w = 4$

$C_5 = 0001100 \rightarrow h_w = 2$

$\therefore \boxed{d_{min} = 2}$



Single Error Correcting Hamming Codes

⑦

we have the parity check matrix as $H = [P^T | I_{n-k}]$

$$\therefore H^T = \begin{bmatrix} -P \\ I_{n-k} \end{bmatrix} \rightarrow \textcircled{1}$$

- From ① we have $(n-k)$ number of columns.
 - In each row of H^T , it has $(n-k)$ number of entries each of which could be '0' or a '1'. Thus we can have 2^{n-k} numbers of distinct rows. But a row of 0's cannot be used as this represents the syndrome of no error. Thus we are left with $[2^{n-k} - 1]$ numbers of distinct rows.
- $\therefore$ The condition for all the rows of H^T to be distinct & it is written as

$$2^{n-k} - 1 \geq n \rightarrow \textcircled{2}$$

$$2^{n-k} \geq n+1$$

$$n-k \geq \log_2(n+1)$$

$$\text{or } k \leq n - \log_2(n+1) \rightarrow \textcircled{3}$$

Thus, the single error correcting (n, k) hamming code has the following parameters.

code length : $n \leq 2^{n-k} - 1$
No of message bits : $k \leq n - \log_2(n+1)$
No of parity check bits : $(n-k)$
Error correcting capability : $t = \frac{d_{min} - 1}{2}$

Hamming Bound : "If an (n, k) Linear block code is to be capable of correcting upto 't' errors, then the total no of syndromes shall not be less than the total no of all possible error patterns.

$$\text{i.e. } 2^{n-k} \geq \sum_{i=0}^t nC_i \rightarrow \textcircled{1}$$

equn ① is referred to as the "Hamming Bound".

A Binary code for which the hamming bound turns out to be an equality is called a "perfect code".

Problems.

- ① Consider the code vectors $C_1 = 10010$, $C_2 = 01101$, $C_3 = 11001$.
find (i) $d(C_1, C_2)$ (ii) $d(C_1, C_3)$ (iii) $d(C_2, C_3)$ and prove that
 $d(C_1, C_2) + d(C_2, C_3) \geq d(C_1, C_3)$

→ (i) $d(C_1, C_2) \Rightarrow C_1 = 10010$
 $C_2 = 01101$

$$d(C_1, C_2) = 5$$

(ii) $d(C_1, C_3) = 3$ (iii) $d(C_2, C_3) = 2$

$$\therefore d(C_1, C_2) + d(C_2, C_3) \geq d(C_1, C_3)$$

$$5 + 2 \geq 3 = \boxed{7 \geq 3.}$$

- ② For a (6,3) linear Block code, the 3 parity check bits are formed from the following equations $C_4 = d_1 \oplus d_3$, $C_5 = d_1 \oplus d_2 \oplus d_3$, $C_6 = d_2 \oplus d_3$.

write 'G' Suppose that the received word is 111010
decode the transmitted code word by finding the location of the error and transmitted data bits.

→ $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow G = \begin{bmatrix} 1 & 0 & 0 & | & 1 & 1 & 0 \\ 0 & 1 & 0 & | & 0 & 1 & 1 \\ 0 & 0 & 1 & | & 1 & 1 & 1 \end{bmatrix}$

given $R = 111010$

$$\therefore S = RH^T$$

$$H^T = \begin{bmatrix} P \\ \vdots \\ I_{n-k} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = 000$$

No error since the syndrome is zero.
The transmitted and received bits are same.

- ③ Consider a (6,3) Linear Block code whose generator matrix is given below (i) find all the code words for this code.
 (ii) Minimum weight of the code. Given $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$
 (iii) Hamming distance.

→ $G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$

2^K code words can be determined

$\therefore 2^3 = 8$ code words (D_1 to D_8)

$\therefore C = [d_1 \ d_2 \ d_3 \ d_1+d_3 \ d_2+d_3 \ d_1+d_2]$

Message	codewords	minimum weight	Hamming distance
000	000000	0	3
001	001110	3	4
010	010011	3	3
011	011101	4	3
100	100101	3	3
101	101011	4	4
110	110110	4	3
111	111000	3	3

- ④ The parity check bits for (8,4) linear Block codes are generated by
 $C_5 = d_2 \oplus d_3 \oplus d_4$
 $C_6 = d_1 \oplus d_2 \oplus d_4$
 $C_7 = d_1 \oplus d_2 \oplus d_3$
 $C_8 = d_2 \oplus d_3 \oplus d_4$
 where d_1, d_2, d_3, d_4 are message bits
 find (i) G
 (ii) H
 (iii) Minimum weight of the code

→ $P = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

(i) $G = [I_k \ P]_{k \times n} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$

$$(ii) H = [P^T \mid I_{n-k}]$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(iii) C = DG \rightarrow 2^4 = 16 \text{ codewords} \quad c = [d_1, d_2, d_3, d_4, d_4 + d_2 + d_3, d_1 + d_2 + d_4, d_1 + d_2 + d_3, d_1 + d_3 + d_4]$$

Message	Codewords	minimum weight.
0000	00000000	0
0001	00011101	4
0010	00101011	4
0011	00110110	4
0100	01001110	4
0101	01011011	4
0110	01100101	4
0111	01110100	4
1000	10000111	4
1001	10011010	4
1010	10101100	4
1011	10110001	4
1100	11001001	4
1101	11010100	4
1110	11100010	4
1111	11111111	8

$$d_{min} = 4$$

✖

⑤ Design an (n, K) hamming code with a $d_{min} = 3$ & message length = 4 bits. Also show that it can correct single errors.

→ Given: $K = 4$, $d_{min} = 3$

By trial & error, the minimum value of code length n which satisfies the inequality is $n \leq 2^{n-k} - 1$

$$\therefore \text{if } n=5, 5 \leq 2^{5-4} - 1 \quad \times$$

$$n=6, 6 \leq 3 \quad \times$$

$$n=7, 7 \leq 7 \quad \checkmark$$

$\therefore n=7$ (minimum value)

$\therefore (7,4)$ LBC

To choose H^T ,

we know $H^T = \begin{bmatrix} P \\ I_{n-k} \end{bmatrix} = \begin{bmatrix} P \\ I_3 \end{bmatrix}$

$$H^T = \begin{bmatrix} [P] \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

'P' is chosen such as way that ① H^T should not contain 'row' of 0's. ② No two rows of H^T must be same i.e all 7 rows of H^T must be distinct.

$$\therefore 2^{n-k} = 2^{7-4} = 2^3 = 8 \text{ combinations are there}$$

they are 000 $\rightarrow$ it cannot be used.

001 $\checkmark$

010 $\checkmark$

011 $\checkmark$

100 $\checkmark$

101

110

111

cannot be used as they are already present as the row of unit matrix.

$\therefore$ only four combinations are left.

011, 101, 110, 111 is chosen & they can be arranged in any order of $[P]$.

$$\therefore [P] = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore H^T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore [G] = [I_k | P] \\ = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\therefore [C] = [D] [G]$$

$$[C] = \{d_1, d_2, d_3, d_4, d_2 + d_3 + d_4, d_1 + d_3 + d_4, d_1 + d_2 + d_4\}$$

Message vector [D]	code-vector [C]	Message vector [D]	code-vector [C]
0000	00000000	1000	10000011
0001	00011111	1001	10011000
0010	00101110	1010	10101011
0011	00110001	1011	10110100
0100	01001011	1100	11001110
0101	01010100	1101	11010011
0110	01100011	1110	11100000
0111	01111000	1111	11111111

To show that it correct single error $d_{min} = 3$

$$\therefore t = \frac{d_{min} - 1}{2} = \frac{3 - 1}{2} = 1$$

So let the received vector with single error be $R = [111100]$

$$\therefore S = RH^T = [111100] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$S = [110]$ which is present in 3rd row

So we need to correct 3rd bit of $R \therefore R = [110100]$

(2) Consider (7,4) hamming code, with $d_{min} = 3$, show that it is a perfect code.

→ To find it is a perfect code
we have $2^{n-k} \geq \sum_{i=0}^t \binom{n}{i}$

$$\text{where } t = \frac{d_{min} - 1}{2} = \frac{3 - 1}{2} = 1$$

$$\therefore 2^{(7-4)} \geq \sum_{i=0}^1 \binom{7}{i}$$

$$2^3 \geq \binom{7}{0} + \binom{7}{1}$$

$$2^3 \geq 1 + 7$$

$$8 \geq 8$$

$$\therefore \boxed{8 = 8}$$

$\therefore (7,4)$ is a perfect code.

(Syndrome Decoding)

Table look up Decoding Using the standard Array

Let $C_i = C_1, C_2, C_3 \dots C_{2^K}$ be the 2^K distinct code vectors of a (n, K) Linear Block code.

Let 'E' be any error pattern.

$$\therefore E_i = E + C_i \rightarrow (1)$$

Where the set of vectors E_i is called the "CO-SET" of the code.

Multiplying equation (1) by H^T on both the sides

$$E_i H^T = E H^T + C_i H^T \quad \text{where } C H^T = 0$$

$$\therefore \boxed{E_i H^T = E H^T}$$

Each element in the standard array is distinct.

Steps to Construct the standard Array.

- (i) 2^K valid code-vectors of the codes are placed in a row with all zero code-vector as first element
- (ii) From the remaining an error E_2 is chosen and is placed below all zero-code vector. The second row can now be formed by placing $E_2 + C_i$ where $i=2, 3, \dots, 2^K$ under C_i
- (iii) Now an unused E_3 is taken & the 3rd row is completed as given in step (ii)
- (iv) The process is continued till all the combination of codes are used. The resulting array for a (n, K) linear block code is shown below.

$C_1 = \text{all } 0\text{'s}$	C_2	C_3	$\dots$	C_{2^K}
E_2	$C_2 + E_2$	$C_3 + E_2$	$\dots$	$C_{2^K} + E_2$
E_3	$C_2 + E_3$	$C_3 + E_3$	$\dots$	$C_{2^K} + E_3$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$E_{2^{n-K}}$	$C_2 + E_{2^{n-K}}$	$C_3 + E_{2^{n-K}}$	$\dots$	$C_{2^K} + E_{2^{n-K}}$

Properties of Standard Array.

- ① the standard array consists of 2^n codewords which are partitioned into 2^k disjoint cosets as shown
- ② the first entry of each row or coset is known as the coset leader & it represents a correctable error pattern for a particular code.
- ③ Each element in the array is distinct. Therefore different columns in array are disjoint.
- ④ Generally error patterns of smallest weight are chosen as coset leaders and decoding which results from, such an array is known as min distance decoding or maximum likelihood decoding. Such an array can be used to correct all such errors induced by the channel only if these errors correspond/match with the coset leaders.
- ⑤ All 2^k error vectors in a single row will have the same syndrome.

Problems

1. Construct the standard array for a (6,3) LBC given $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Detect and correct errors for the received vectors $R_1 = 100100$ and $R_2 = 000011$. Draw the error correction for the same.

→ $C = [D] [G]$, we have 2^k code vectors $\rightarrow 2^3 = 8$

$$[G] = [I_k | P] = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \quad H^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore C = [d_1 \ d_2 \ d_3] \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$C_1 = d_1 \quad C_2 = d_2 \quad C_3 = d_3 \quad C_4 = d_1 + d_3 \quad C_5 = d_2 + d_3 \quad C_6 = d_1 + d_2 + d_3$$

D	[C]
000	0000000
001	0011111
010	0100111
011	0111000
100	1001011
101	1010100
110	1101100
111	1110011

Standard Array

Syndrome	Coset leader	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇
000	000000	001111	010011	011100	100101	101010	110110	111001
101	100000	101111	110011	111100	000101	001010	010110	011001
011	010000	011111	000011	001100	110101	111010	100110	101001
111	001000	000111	011011	010100	101101	100010	111110	110001
100	000100	001011	010111	011000	100001	101110	110010	111101
010	000010	001101	010001	011110	100111	101000	110100	111011
001	000001	001110	010010	011101	100100	101011	110111	111000
110	110000	111111	100011	101100	010101	011010	000110	001001

⑥

$$S = RH^T$$

$$= [100100] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

match the S with H^T , it is present in 6th row.

$\therefore$ 6th Bit is in error

$$\therefore \text{corrected code vector} = [100101]$$

$$S = [001]$$

Also match the R_1 with the elements of table. The Received Vector R_1 is present in the 4th column of 7th row. hence the corrected code vector corresponds to the entry in top most row of the column 4 i.e 100101

(c) Error correction circuit.

$$S_1 = \eta_1 + \eta_3 + \eta_4$$

$$S_2 = \eta_2 + \eta_3 + \eta_5$$

$$S_3 = \eta_1 + \eta_2 + \eta_3 + \eta_6$$

$$\therefore e_1 = S_1 \bar{S}_1 S_2$$

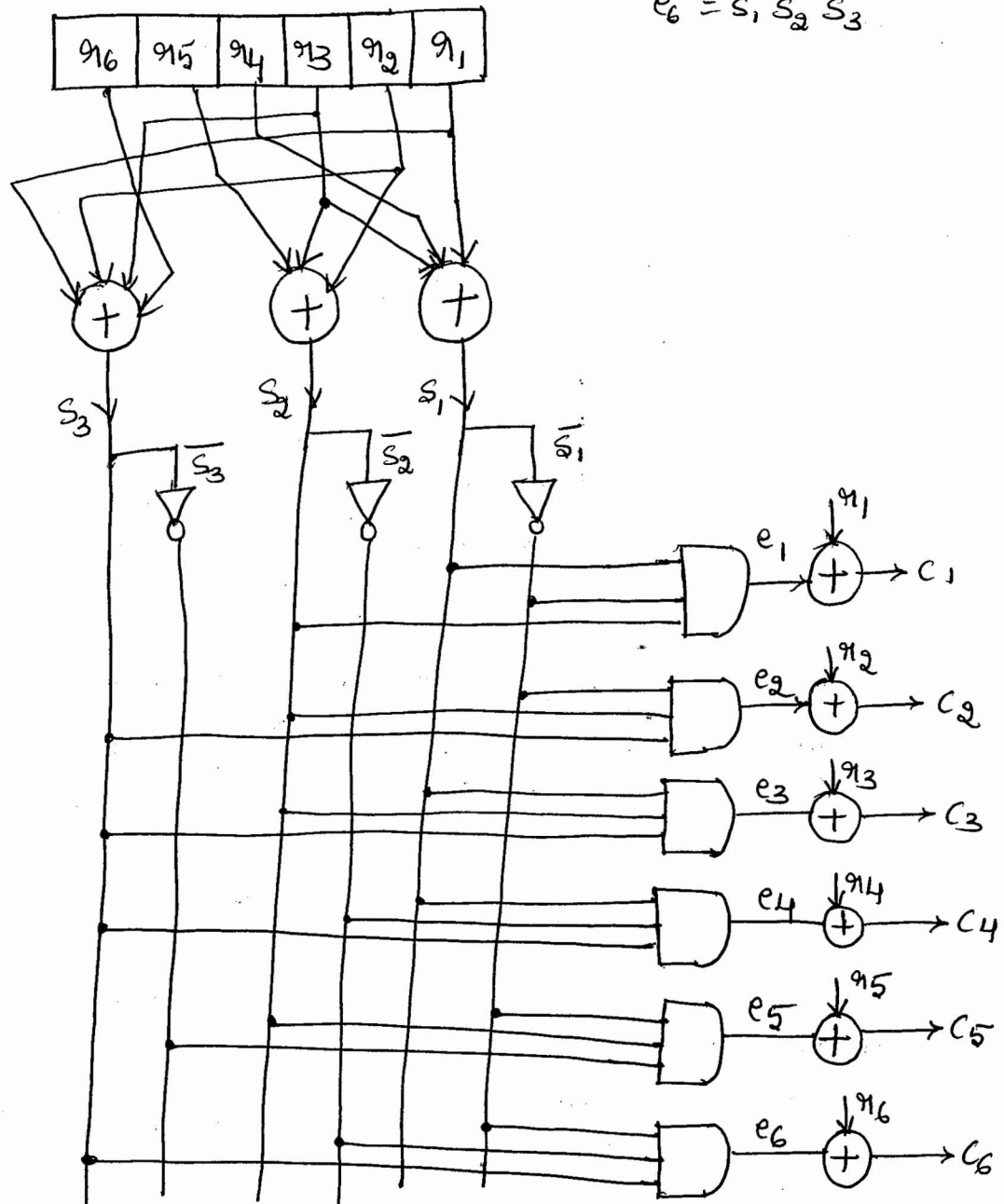
$$e_2 = \bar{S}_1 S_2 S_3$$

$$e_3 = S_1 S_2 S_3$$

$$e_4 = S_1 \bar{S}_2 S_3$$

$$e_5 = \bar{S}_1 S_2 \bar{S}_3$$

$$e_6 = \bar{S}_1 \bar{S}_2 S_3$$



② For a (5,2) LBC, the generator matrix is of the form $[I_k | P]$ where $[P]$ is given by $[P] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

find (a) generator Matrix (b) Parity Check matrix (c) all possible valid code-vectors (d) d_{min} and error correcting capability of the code (e) Construct standard array by listing all the 32 five-tuple patterns.

→ (a) Generator matrix is given by $[G] = [I_k | P] = [I_2 | P]$

$$[G] = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

(b) The parity check matrix will be of the form $[P^T | I_{n-k}]$ given by $[H] = [P^T | I_{5-2}] = [P^T | I_3]$

$$[H] = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(c) Code-Vectors are generated Using

$$[C] = [D] [G] = [d_1 \ d_2] \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$= [d_1, d_2, (d_1 + d_2), (d_1 + d_2), d_1]$$

Message Vector	Code-Vector	H_{10}
00	00000	0
01	01110	3
10	10111	4
11	11001	3

(d) the lowest hamming weight of a non-zero code vector is 3

$$t = \frac{d_{min} - 1}{2} = \frac{3 - 1}{2} = 1$$

∴ (5,2) code is a SEC code

(e) Standard array

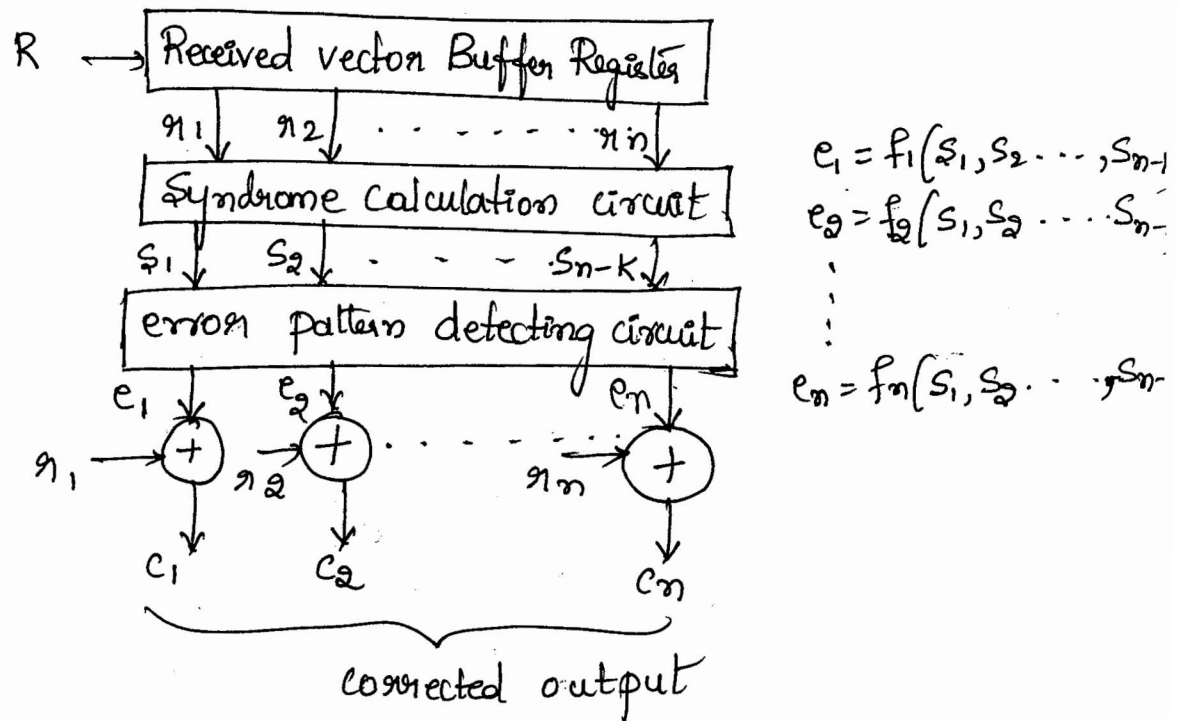
Syndrome	coset leader			
000	00000	01110	10111	11001
111	10000	11110	00111	01001
100	01000	00110	11111	10001
000	00100	01010	10011	11101
000	00010	01100	10101	11011
001	00001	01111	10110	11000
001	00011	01101	10100	11010
101	00101	01011	10010	11100

General Decoding circuit for (n, k) LBC.

The implementation of the decoding circuit (error correcting circuit), the standard array can be regarded as the truth table of n -switching functions.

The Truth Table consists of 2 columns

- ① the syndrome column
- ② coset leader column



Where $s_1, s_2, \dots, s_{n-k}$ are syndrome digits
 $e_1, e_2, \dots, e_n$ are error digits of coset leaders.

for Problem ① in page ⑩.

→ Draw the complete error correcting circuit

$$S = [s_1, s_2, s_3] = [r_1 + r_3 + r_4, r_2 + r_3 + r_5, r_1 + r_2 + r_6]$$

The error bits of the coset leader can be expressed in terms of syndrome bits.

$$\therefore e_1 = s_1 \bar{s}_2 \bar{s}_3 \quad e_2 = \bar{s}_1 s_2 s_3 \quad e_3 = s_1 s_2 \bar{s}_3 \quad e_4 = s_1 \bar{s}_2 s_3$$

$$e_5 = \bar{s}_1 s_2 \bar{s}_3 \quad e_6 = \bar{s}_1 \bar{s}_2 s_3$$

Karnaugh correcting circuit can be written as:-

